Course Map

Claude edition — make the claim earn its place

Ten investigations, joined by evidence

ONE BOUNDARY FOR THE WHOLE COURSE

Learner investigations use isolated batteries at 12 V DC or less. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. The adult facilitator controls materials and stop decisions; that does not make the adult electrically qualified. Lesson 10 and the final machine are explanatory unless the work is performed by a person qualified for that work.

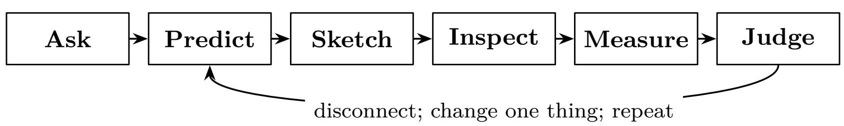
The bargain

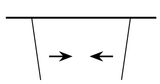
You are not asked to believe an effect because it has a famous name. You make a prediction, arrange a bounded test, and keep the numbers that answer the question. A failed prediction is useful. A dramatic result with no controls is not. By the end, every arrow in the machine diagram must point back to a measurement, a calculation, or an explicit limit.

Predict first

Before Lesson 1: Which would persuade you most — one large meter deflection, five smaller deflections that follow distance, or a formula in a book? Write what the other two pieces of evidence would add.

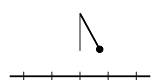
One complete investigation





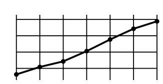
Name the fork

What two outcomes would mean different things? Commit before preparing the apparatus.



Read a quantity

A number needs a unit, instrument range, geometry, and conditions or it cannot be repeated.



Earn the pattern

Repetition shows whether the response outruns scatter, drift, and wishful thinking.

ENTRY CHECK

The learner points to the disconnect, states the voltage limit, names the stop conditions, and writes a prediction that could fail. Otherwise begin with the Safety sheet, not the apparatus.

Course Map

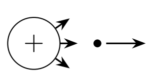
Claude edition — a staircase, not a box of tricks

Ten investigations, joined by evidence

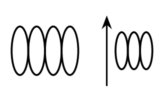
What each lesson must leave behind

#	Question	Evidence	Ready-to-advance test	Notebook artifact
1	Charge	Repeated attraction/repulsion comparisons and can motion versus distance.	Explain charge transfer without confusing it with polarization.	A sign matrix and distance notes.
2	Fields	Test-object direction and deflection at five or more positions.	Read an arrow as force on a positive test charge, not a visible strand.	Measured field map.
3	Ohm's law	Matched voltage/current pairs for one fixed resistance.	Decide whether V/I is stable within meter resolution.	Table, graph, worked ratio.
4	Electromagnets	Response versus current or turns, with the other quantities fixed.	Name which of current, turns, core, and geometry actually changed.	One-variable comparison.
5	Induction	Polarity and size versus motion direction or speed.	Show that the change, not mere proximity, predicts the voltage.	Frame-by-frame motion record.
6	Transformers	Turn counts and both measured voltages.	Use the ratio to predict direction and scale, then explain loss.	Ratio and discrepancy budget.
7	Capacitors	Low-voltage charge/discharge readings at fixed times.	Distinguish stored energy from a source still connected.	Time series with discharge check.
8	Resonance	Response throughout a controlled low-energy frequency sweep.	Show that a repeatable band exceeds noise, not merely one peak reading.	Sweep graph and peak interval.
9	Coupling	Received response versus distance or orientation.	Use geometry to predict a reversible change in transfer.	Scale sketch plus readings.
10	Spark gap	Source-based explanation of gas breakdown and switching; no learner energization or build.	Explain exactly why this mechanism crosses the learner boundary.	Mechanism and hazard map.

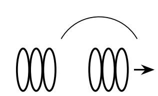
The three chains



charge → field → force
Lessons 1–2



current → field → induction
Lessons 3–5



ratio → tuning → coupling
Lessons 6–9

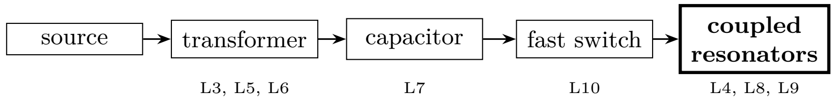
PROOF POINT A — AFTER LESSON 3

Draw a complete measuring circuit from memory. Mark source polarity, conventional current, voltage-meter placement, current-meter placement, and the expected scale of both readings. Then compare drawing and bench.

PROOF POINT B — AFTER LESSON 6

Predict what doubling secondary turns changes. Give the ideal ratio, then name two real mechanisms that can pull a measurement away from it.

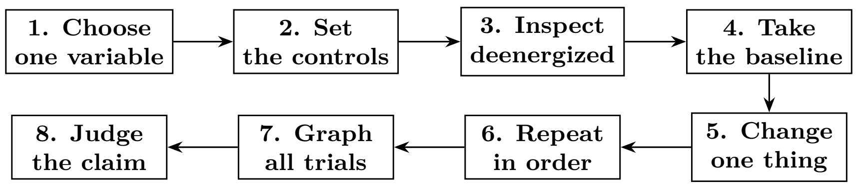
The final machine is a model, not a learner project



PROOF POINT C — AFTER LESSON 9

Trace energy through the diagram without saying "it just steps up." At every arrow, name a measurable quantity and one observation that would expose a broken link. This is the prerequisite for reading the final explanatory build.

The measurement discipline



Plan before wiring

Question	What relationship will the apparatus decide?
Prediction	If ______________ changes, then ______________, because ________________.
Changed quantity	Planned values and unit.
Response	Instrument, mode, range, resolution, and unit.
Controls	Parts, source, geometry, timing, preparation, environment.
Stop	Heat, odor, pain, damage, loose connection, unstable reading, surprise, or facilitator decision.

Raw trials — acquisition order, never tidied order

Trial	Changed quantity + unit	Response + unit	Controls checked	Observation / anomaly
1				
2				
3				
4				
5				

Four honest verdicts

Supported. Repetition follows the predicted direction and the effect is comfortably larger than resolution and ordinary scatter.

Needs a sharper model. The direction works but the size misses. Audit ideal assumptions, loading, resistance, geometry, timing, and uncertainty.

Not supported. The predicted pattern is absent or reverses. Keep the result. Check the test, then change the explanation.

Inconclusive. A control drifted, the response sits inside noise, or inspection was incomplete. Improve the test before making a claim.

Course Map

Claude edition — the notebook is part of the apparatus

Ten investigations, joined by evidence

WHEN RESULTS DISAGREE

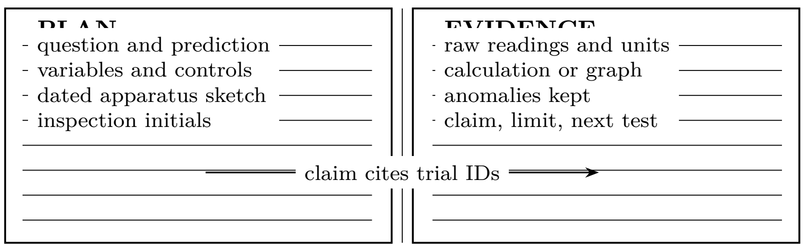
Disconnect. Compare the drawing with the apparatus. Confirm meter leads, mode, range, zero, and resolution. Inspect every joint. Re-run a known baseline. Do not increase the source or leave the course boundary to force a visible result.

Predict first

Questions worth asking aloud: What observation would change your mind? What else could cause this reading? Which control is most fragile? Is the change larger than the instrument can resolve? Could a second learner repeat the test from the record without asking you what you meant?

A record another person can audit

Use a bound notebook and number every page. Start each lesson on a clean spread. Never tear out an inconvenient trial. Give every figure and table an ID such as `L05-T1`; revisions point back to earlier IDs rather than replacing them.



Every lesson record

Header	Date, lesson, people, apparatus ID, source voltage, instrument and range.
Question	One answerable relationship or mechanism.
Prediction	Direction, approximate magnitude when possible, and reason.
Plan	Changed quantity, response, controls, sequence, and stop rule.
Sketch	Deenergized wiring or geometry with values, polarity, meter placement, distances, and orientation.
Raw evidence	All readings in order, units in headings, resolution, anomalies.
Analysis	Graph or calculation, comparison with prediction, uncertainty.
Claim	One bounded conclusion tied to named trials.
Revision	What changed in the model; unresolved question; discriminating next test.
Sign-off	Learner check, facilitator inspection, boundary maintained.

Course index

L.	Record IDs	Measured pattern	Model change	Open question / next link
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Course Map

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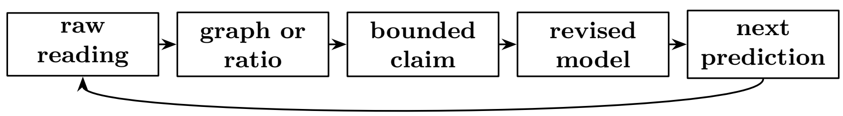
The course is complete when the evidence reconnects

Three claim audit

Choose one claim from Lessons 1–3, one from 4–6, and one from 7–9. For each:

1. locate the raw readings and the deenergized apparatus drawing;
2. reproduce one calculation or graph without copying its conclusion;
3. state the largest uncertainty and one plausible alternative mechanism;
4. name a result that would weaken the claim;
5. propose the smallest safe test that distinguishes the explanations.



FINAL ORAL PROOF

With the machine diagram covered, reconstruct the chain from source to coupled resonators. At every step state: the physical change, the quantity that shows it, the lesson that measured it, and the safety boundary. Then uncover the diagram and audit omissions. "High voltage happens" is not an explanation.

Demonstrations are evidence only when recorded

Use each demonstration after its named prerequisite. Before it begins, write a prediction and decide what observation bears on it. Afterwards, distinguish what was seen from what was inferred. A demonstration may open a question; it does not replace the learner's bounded measurement.

Independent edition, shared physical world

This edition's prose and finished diagrams are independently authored. The provider-neutral research library supplies authoritative physics and safety claims, with reusable citations and scope notes. Authority still does not replace measurement: it tells us what is already established and where a learner test must stop.

The one-sentence test

A changing source builds and transfers energy through electric and magnetic fields; circuit values and geometry select how much, how fast, and where it goes. The course has done its job only when the learner can attach a measured record and a stated limit to every clause.